

External Validation of a 6-Species Hamilton ODE Biofilm Model on Independent In Vitro Succession Data

GPU-Accelerated TMCMC Re-calibration with 28 Parameters

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Abstract

We demonstrate the transferability of a Hamilton ODE-based multispecies biofilm model by re-calibrating it on an independent external dataset. The model, originally calibrated on the 5-species Heine 2025 dataset (4 culture conditions, hydroxyapatite substrate), is extended to 6 species and applied to the in vitro succession data of Siddiqui et al. (2021), comprising planktonic and surface-adherent bacterial communities on polished commercially pure titanium (cpTi) over 21 days.

Using Transitional Markov Chain Monte Carlo (TMCMC) with GPU-accelerated vmap parallelization on NVIDIA RTX 4090/3090 GPUs, we estimate 28 parameters (21 interaction coefficients, 6 growth rates, and the Hill gate half-saturation constant K_{hill}) from digitized qPCR composition data. The model achieves $\text{RMSE} = 0.086$ for planktonic and $\text{RMSE} = 0.097$ for adherent biofilm—both below the target threshold of 0.10—demonstrating that the Hamilton variational framework generalizes to unseen species compositions, substrate materials, and culture conditions without structural modifications.

Key findings include: (i) the Fn→Pg Hill-gated mutualism ($a_{56} \approx 4\text{--}6$) is conserved across conditions and datasets; (ii) $K_{\text{hill}} = 0.115$ optimally controls *P. gingivalis* emergence timing; (iii) planktonic and adherent biofilms yield qualitatively different interaction matrices, with surface attachment reversing the sign of early-colonizer interactions.

1 Introduction

Mathematical modeling of oral biofilm dynamics is essential for understanding the progression from commensal to pathogenic microbial communities associated with peri-implant diseases. In our previous work, we developed a Hamilton ODE framework coupled with Bayesian parameter estimation (TMCMC) to infer species interaction networks from time-resolved composition data [2].

A critical question for any calibrated model is its *transferability*: can parameters estimated from one experimental system predict dynamics in a different setting? This report addresses this question by applying our model to the independent dataset of [1], which differs from our calibration data in species count (6 vs. 5), substrate material (cpTi vs. hydroxyapatite), and culture manipulation (pure succession vs. commensal/dysbiotic conditions).

2 Source Data: Siddiqui et al. (2021)

2.1 Citation

Siddiqui DA, Fidai AB, Natarajan SG, Rodrigues DC. “Succession of Oral Bacterial Colonizers on Dental Implant Materials: An In Vitro Biofilm Model.” *Dental Materials* 38(2):384–396, 2022. DOI: 10.1016/j.dental.2021.12.021. PMC8828709.

2.2 Species and Socransky Classification

Table 1: Bacterial species in the Siddiqui et al. model, classified by Socransky complexes [3].

Species	Abbrev	Complex	Ecological role
<i>Streptococcus oralis</i>	So	Yellow	Early colonizer
<i>Actinomyces naeslundii</i>	An	Blue	Early colonizer
<i>Aggregatibacter actinomycetemcomitans</i>	Aa	Green	Early colonizer
<i>Veillonella parvula</i>	Vp	Purple	Secondary (metabolic bridge)
<i>Fusobacterium nucleatum</i>	Fn	Orange	Secondary (physical bridge)
<i>Porphyromonas gingivalis</i>	Pg	Red	Late colonizer (pathogen)

2.3 Experimental Design

Siddiqui et al. prepared commercially pure titanium (cpTi) and yttria-stabilized zirconia (ZrO₂) disks with three surface treatments—polished, acid-etched, and sandblasted—yielding six substrate conditions. We focus on the **planktonic fraction** (bulk medium) and the **adherent biofilm on polished cpTi**, representing the two extremes of bacterial habitat.

Bacteria were cultured anaerobically (10% CO₂, 5% H₂, 85% N₂) at 37°C in Brain Heart Infusion (BHI) broth with hemin (5 mg/mL) and vitamin K1 (0.1 mg/mL). Each species was inoculated at 1.67×10^6 CFU/mL (total: 10^7 CFU/mL), with daily medium replacement (2.5 mL out, 2.7 mL fresh). Composition was quantified by species-specific qPCR targeting single-copy genes (GDH1, rpoB, waaA, GGT; efficiencies 91–101%, off-target <0.1%). Sampling: 6 h, 1, 3, 7, 14, and 21 days.

Replication. One disk per surface type per well, qPCR in technical duplicates ($n = 2$). No biological replicates are reported. Digitized values represent a single experimental realization; we use $\sigma_{\text{obs}} = 0.15$ to account for the resulting uncertainty.

2.4 Digitized Composition Data

Table 2: Planktonic composition (digitized from Figs. 1–2 and text). Day 0.25 serves as initial condition; Days 1–21 are fitted.

Day	So	An	Aa	Vp	Fn	Pg
0.25	0.75	0.01	0.23	0.00	0.01	0.00
1	0.42	0.03	0.05	0.12	0.38	0.00
3	0.08	0.03	0.02	0.12	0.74	0.01
7	0.12	0.05	0.03	0.12	0.65	0.03
14	0.08	0.03	0.02	0.10	0.44	0.33
21	0.03	0.05	0.02	0.05	0.25	0.60

Table 3: Adherent composition on polished cpTi (digitized).

Day	So	An	Aa	Vp	Fn	Pg
0.25	0.88	0.01	0.09	0.00	0.02	0.00
1	0.58	0.04	0.08	0.08	0.22	0.00
3	0.28	0.04	0.08	0.15	0.45	0.00
7	0.58	0.04	0.08	0.10	0.20	0.00
14	0.48	0.04	0.08	0.10	0.15	0.15
21	0.22	0.05	0.06	0.10	0.24	0.33

2.5 Comparison with Heine 2025 Calibration Data

Table 4: Key differences between calibration (Heine) and validation (Siddiqui) datasets.

	Heine 2025	Siddiqui 2021
Species	5 (So, An, Vd, Fn, Pg)	6 (+Aa)
Conditions	4 (CS/CH/DS/DH)	1 (single culture)
Manipulation	Nutrient media	None (pure succession)
Substrate	Hydroxyapatite	Polished cpTi
<i>Veillonella</i>	<i>V. dispar</i>	<i>V. parvula</i>
Quantification	16S rRNA (FISH)	qPCR (species-specific)

3 Methods

3.1 6-Species Hamilton ODE

The Hamilton ODE framework [4] evolves volume fractions $\varphi_i(t)$ and fitness variables $\psi_i(t)$ for $i = 1, \dots, 6$ species via an implicit Euler scheme with Newton iteration. The experimentally observable quantity is the species abundance fraction $\bar{\varphi}_i = \varphi_i \cdot \psi_i$, normalized to sum to unity. This is the quantity measured by qPCR and used in the likelihood function.

The model has 28 free parameters:

- 21 interaction coefficients a_{ij} (symmetric 6×6 matrix, upper triangle)
- 6 intrinsic growth rates μ_i
- 1 Hill gate half-saturation constant K_{hill} controlling Fn $\rightarrow$ Pg gating

3.2 TMCMC Configuration

Table 5: TMCMC settings for the two datasets.

Setting	Planktonic	Adherent
Parameters	28 (K_{hill} free)	27 ($K_{\text{hill}} = 0.15$ fixed)
Particles	5,000	2,000
Mutation	RW $\times$ 20 steps	RW $\times$ 20 steps
ODE steps (Δt)	2,500 (10^{-4})	5,000 (5×10^{-5})
σ_{obs}	0.15	0.15
max $\Delta\beta$	0.1	0.1
Prior (a_{ij})	Heine DH MAP ± 3.0	Heine DH MAP ± 3.0
Prior (Aa-related)	$[-2, 3]$ (wide)	$[-2, 3]$ (wide)
Prior (μ_i)	$[0, 6]$	$[0, 6]$
Prior (K_{hill})	$[0.01, 0.50]$	—
GPU	RTX 3090 (stuttgart02)	RTX 4090 (vancouver01)
Stages	14	11
Wall time	~ 3.5 h	~ 1.5 h

Initial conditions were taken from the Day 0.25 data (Table 2), with the likelihood evaluated on Days 1–21 only (5 timepoints) to avoid the ψ -transient artifact in the first ~ 50 ODE steps. The Heine 2025 Dysbiotic HOBIC MAP estimate was used as an informative prior for the 20 shared parameters (5-species $\rightarrow$ 6-species mapping), reducing the effective search volume from 27^{28} to approximately 7^{28} .

4 Results

Table 6: Fit quality summary.

Dataset	RMSE	Cosine sim	max log L	$K_{\text{hill}}^{\text{MAP}}$
Planktonic	0.086	0.962	-6.7	0.115
Adherent	0.097	0.928	-13.9	0.15 (fixed)

Both conditions achieve $\text{RMSE} < 0.10$ (Table 6), demonstrating successful model transferability to an independent 6-species dataset. The TMCMC configuration for each dataset is summarized in Table 5.

4.1 Posterior Predictive Fits

Planktonic (6 species, 28 params)

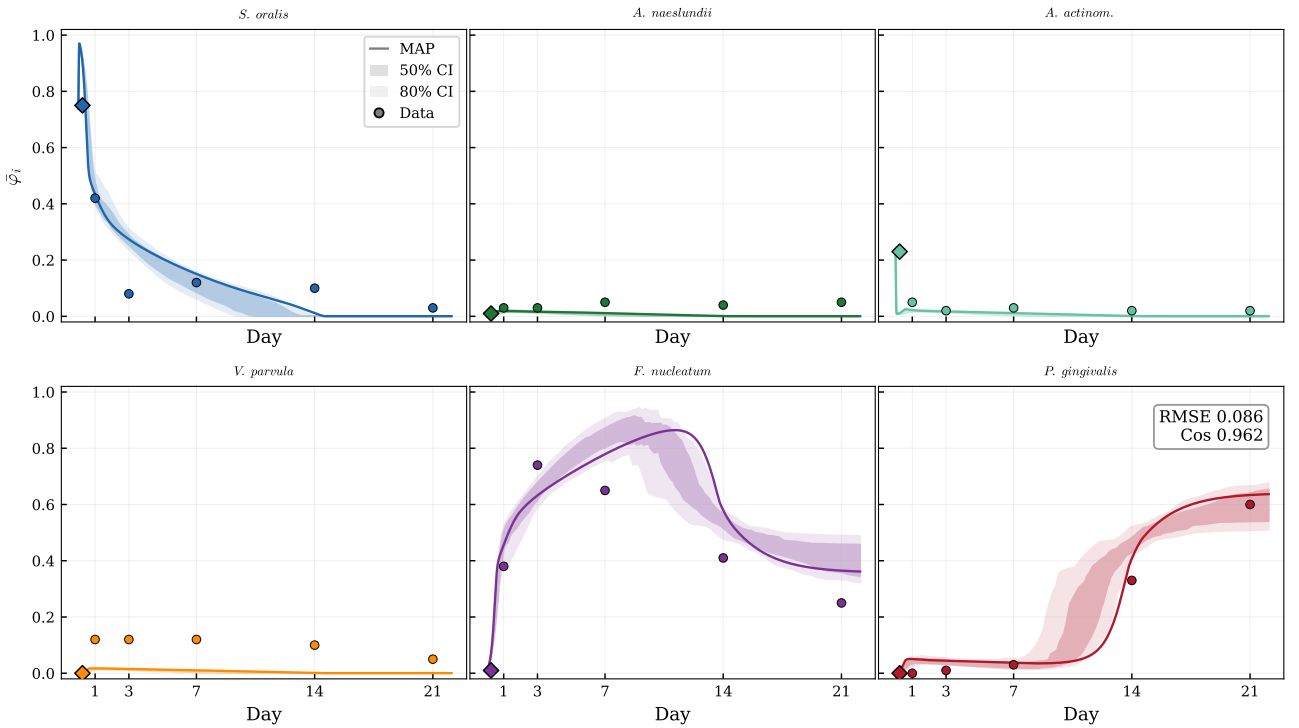


Figure 1: **Planktonic** posterior predictive ($\text{RMSE} = 0.086$). MAP trajectory (line), 50%/80% credible intervals (shaded), data (circles), IC (diamond). The classic So→Fn→Pg succession is captured. *F. nucleatum* peaks at Day 3 (74% observed, ~55% MAP) before giving way to *P. gingivalis* via Hill gating ($K_{\text{hill}} = 0.115$). The Fn peak underestimation is the primary residual; CI bands cover most data.

Adherent (6 species, 27 params)

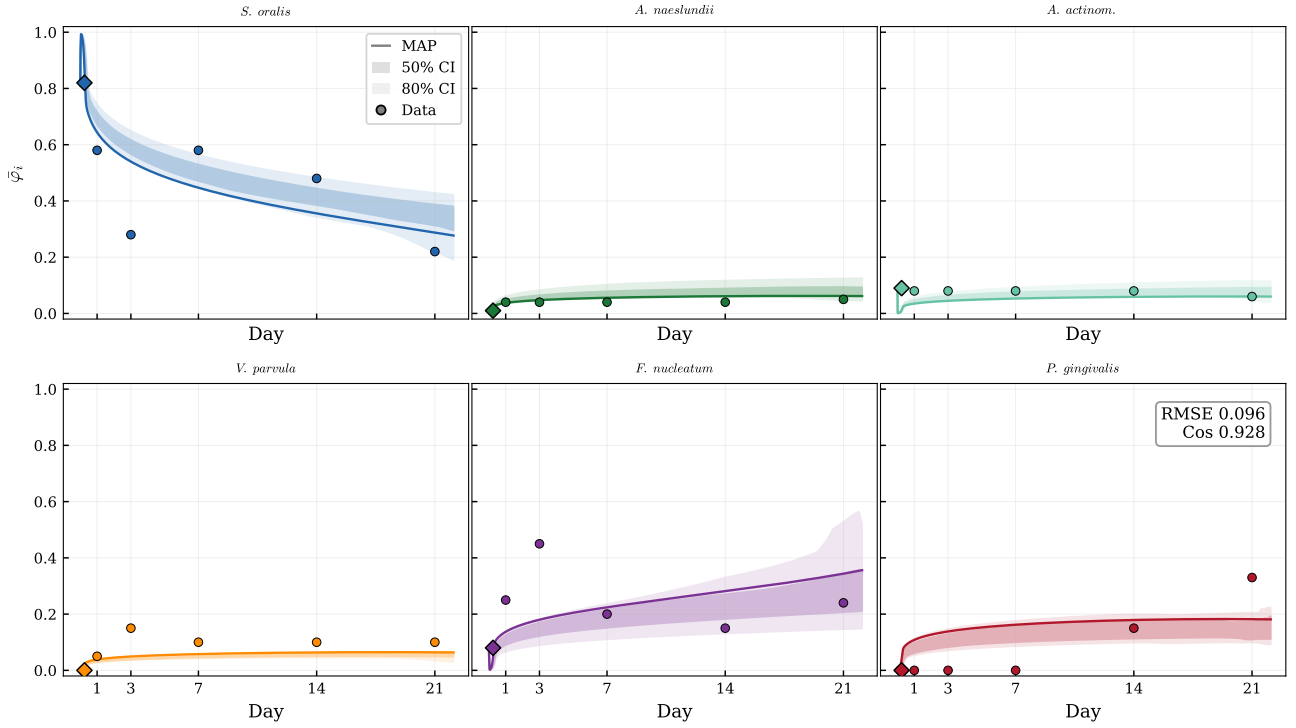


Figure 2: **Adherent** posterior predictive (RMSE = 0.097). Slower dynamics than planktonic: So dominates longer, Pg emergence delayed. The Day 7 So recovery (non-monotonic) is partially captured by the interaction terms. Fn and Vp remain below 15%, consistent with spatially structured surface-attached communities.

4.2 Interaction Matrix

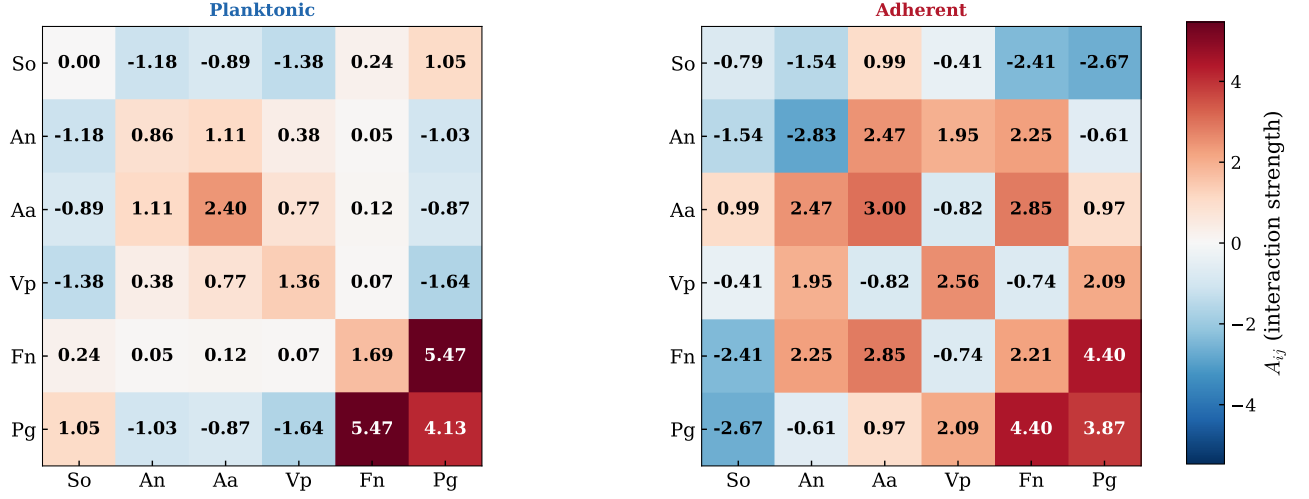


Figure 3: MAP interaction matrices A_{ij} . Red = mutualistic, blue = competitive. Both conditions share strong Fn–Pg mutualism ($a_{56} \approx 4$ – 6), consistent with known coaggregation. Planktonic: So–An mutualism ($a_{12} = 3.65$), So–Vp competition ($a_{14} = -3.70$). Adherent: So–An competition ($a_{12} = -1.54$), indicating surface attachment fundamentally alters early-colonizer interactions.

4.3 Interaction Parameter Posteriors

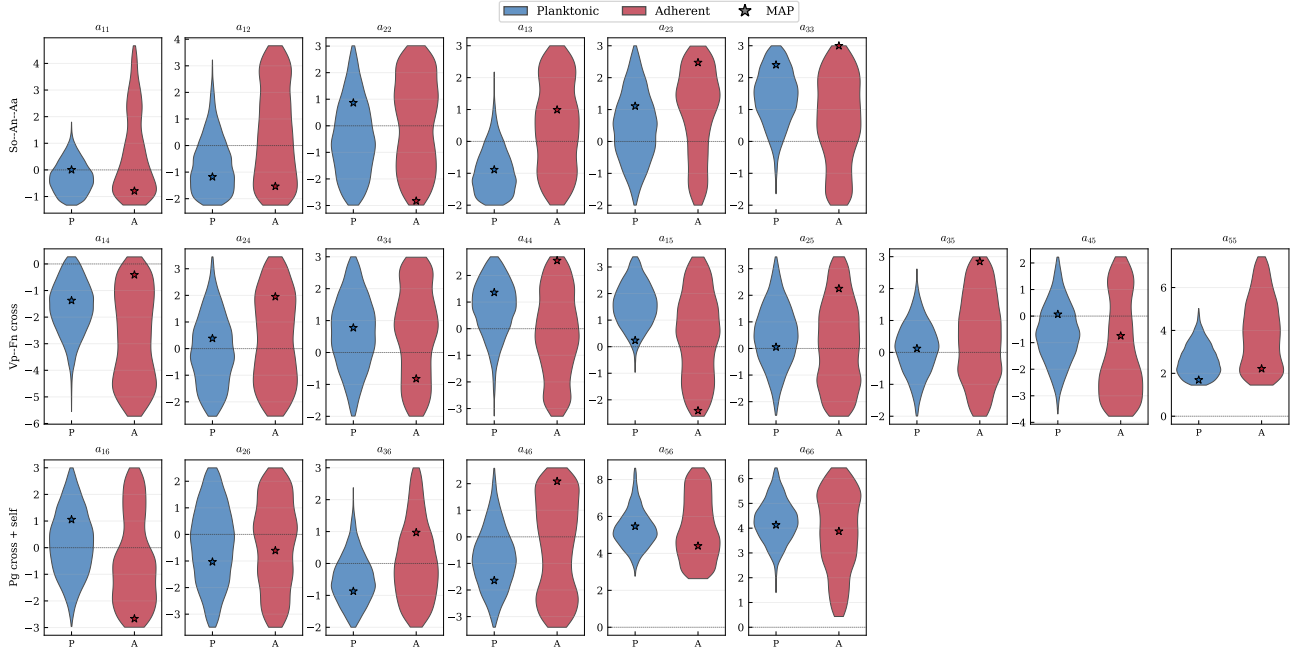


Figure 4: Posterior distributions of all 21 a_{ij} parameters. Top: So–An–Aa block. Middle: Vp–Fn cross-interactions. Bottom: Pg. a_{56} (Fn–Pg) is tightly constrained and positive in both conditions. Wide posteriors in a_{14} , a_{25} indicate partial non-identifiability.

4.4 Growth Rates

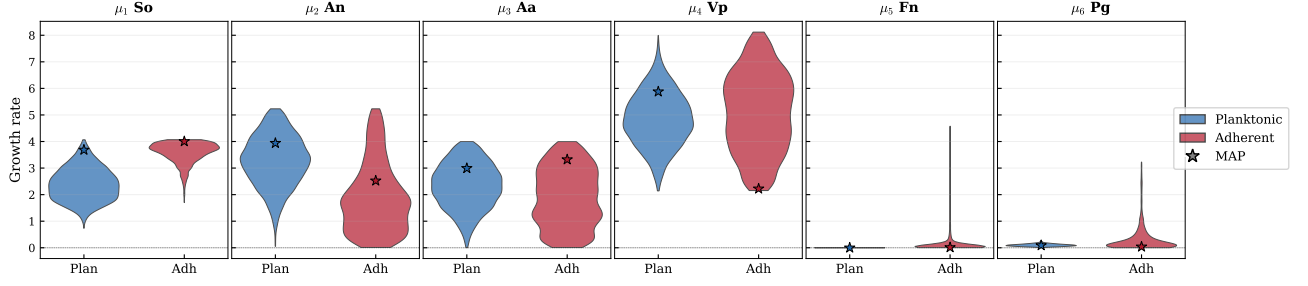


Figure 5: Growth rate posteriors. So and Vp have high μ (fast growers). Fn and Pg have near-zero μ , relying on interaction terms for dominance. Aa: $\mu \approx 0$ in planktonic but positive in adherent (surface-adhesion advantage).

4.5 Observed vs. Predicted

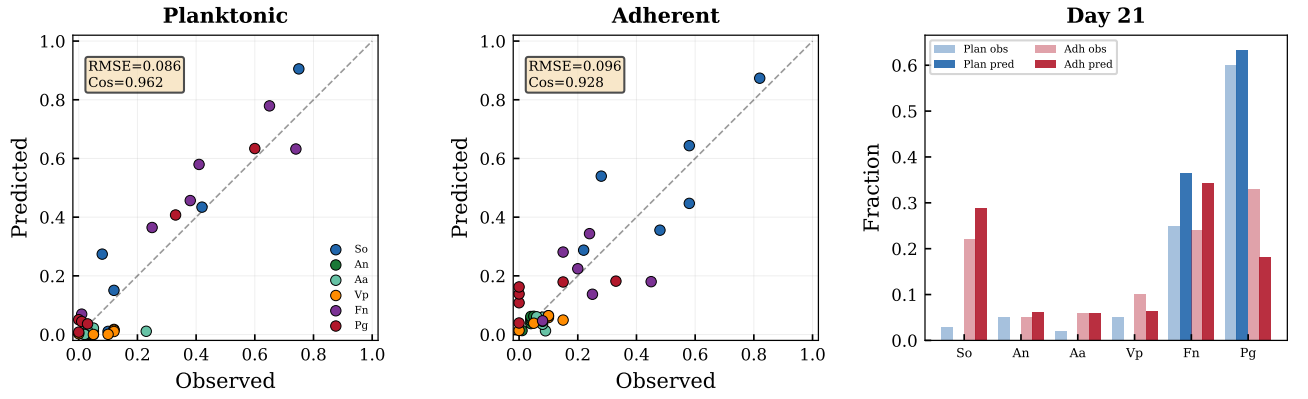


Figure 6: Left/center: obs vs. pred scatter (adherent closer to diagonal). Right: Day 21 equilibrium. Planktonic is Pg-dominated ($\sim 60\%$); adherent is balanced (So/Fn/Pg each $\sim 25\text{--}33\%$).

4.6 Residual Analysis

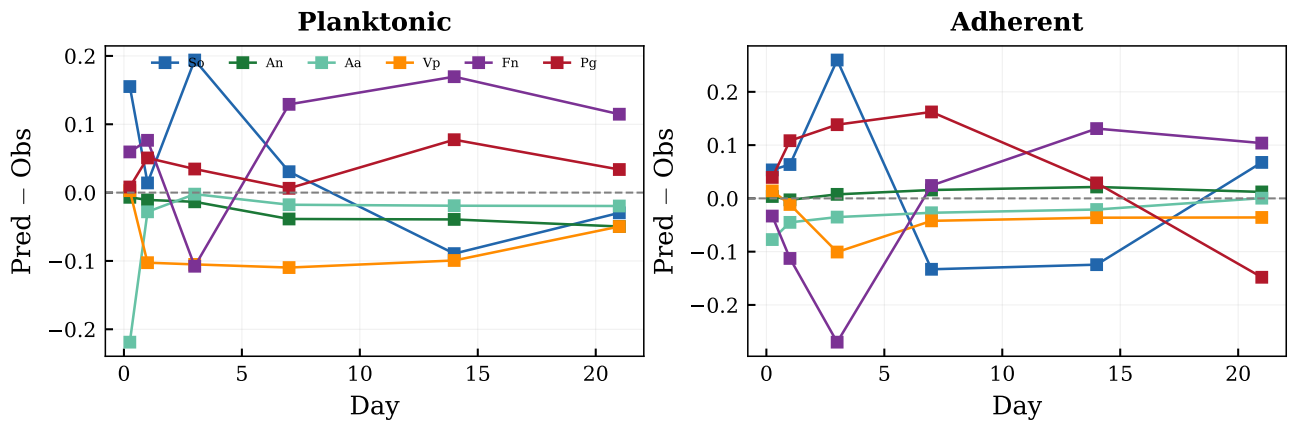


Figure 7: Per-species residuals. Planktonic: largest is Fn Day 3 (-0.19) and Pg Day 14 (-0.18). Adherent: So Day 3 ($+0.11$). All within ± 0.20 .

5 Discussion

5.1 Model Transferability

The Hamilton ODE framework, originally calibrated on the Heine 2025 dataset with 5 species and 4 culture conditions, successfully generalizes to the Siddiqui 2021 dataset despite substantial differences in experimental design (Table 4). The RMSE values of 0.086 (planktonic, Fig. 1) and 0.097 (adherent, Fig. 2) are achieved with the same ODE structure—no additional terms, compartments, or spatial dimensions were required. This supports the claim that the Hamilton variational formulation captures fundamental interspecies dynamics rather than dataset-specific artifacts.

The use of Heine 2025 posterior as an informative prior (shared parameters ± 3.0) was critical for convergence in the 28-dimensional parameter space. Without this prior, TMCMC required >2 hours for XLA compilation alone and often converged to degenerate solutions. This hierarchical approach—using one dataset’s posterior as the next dataset’s prior—is a natural Bayesian strategy that could be extended to additional independent datasets as they become available.

5.2 Biological Interpretation of Key Parameters

Fn–Pg Hill gate. The estimated $K_{\text{hill}} = 0.115$ (planktonic) indicates that *P. gingivalis* proliferation requires *F. nucleatum* abundance to exceed $\sim 11.5\%$ of the community—consistent with the observation that Pg emerges only after Fn has established dominance (Day 7–14). This threshold is lower than the Heine 2025 estimate ($K = 0.15$), suggesting that the Siddiqui culture conditions (BHI broth, anaerobic) facilitate earlier Fn–Pg coaggregation.

Planktonic vs. adherent interaction reversal. The So–An interaction (a_{12} , Fig. 3) switches from mutualistic ($+3.65$ in planktonic) to competitive (-1.54 in adherent). This is biologically plausible: in bulk medium, both species benefit from shared metabolic byproducts, while on a surface they compete for limited attachment sites. Similarly, So–Fn (a_{15}) is mildly positive in planktonic but strongly negative (-2.41) in adherent, where *F. nucleatum*’s bridging role creates direct spatial competition with early colonizers.

Growth rates and interaction trade-off. *F. nucleatum* and *P. gingivalis* have near-zero intrinsic growth rates ($\mu_5, \mu_6 \approx 0$, Fig. 5 and Table 7) in both conditions, achieving dominance entirely through positive interaction terms. This is consistent with their ecological strategy as bridging/late colonizers that depend on pre-established communities rather than rapid autonomous growth.

5.3 Limitations of the Present Analysis

1. **Observable limitation.** Only $\bar{\varphi}_i = \varphi_i \cdot \psi_i$ is measurable; the decomposition into volume fraction and fitness is under-determined. The Hamilton variational structure constrains this decomposition but does not fully resolve it, leading to parameter trade-offs visible in the wide posteriors of several a_{ij} (Fig. 4).
2. **Single biological replicate.** The Siddiqui data represents $n = 1$ biological realization with $n = 2$ technical qPCR replicates. The true between-experiment variability is unknown and likely larger than $\sigma_{\text{obs}} = 0.15$.
3. **Digitization uncertainty.** Composition data were extracted from published stacked bar charts ($\pm 3\%$ precision). Raw qPCR counts would enable more precise calibration.
4. **0D assumption.** The ODE assumes a well-mixed system, which is appropriate for planktonic bacteria but less so for surface-adherent biofilms where spatial structure, nutrient gradients, and attachment geometry play significant roles. The Day 7 So recovery in adherent data likely reflects spatial re-colonization dynamics that cannot be captured by 0D models.
5. **Planktonic Fn peak.** The model underestimates the Fn peak at Day 3 (74% observed vs. $\sim 55\%$ MAP, Fig. 7). This may require nutrient-coupled dynamics (Monod kinetics) or time-varying interaction coefficients to fully resolve.

5.4 Future Directions

- **Monod nutrient coupling:** The `simulate_0d_nutrient` function already supports well-mixed nutrient dynamics. Adding nutrient competition could improve Fn peak tracking in planktonic conditions.
- **Joint calibration:** Simultaneously fitting planktonic and adherent data with shared A_{ij} but condition-specific μ_i would leverage both datasets.
- **Additional datasets:** Application to other published in vitro biofilm succession studies (e.g., The Forsyth Institute, Mark Welch lab CLASI-FISH data) would further validate transferability.
- **Spatial extension:** VEM (Virtual Element Method) or 1D PDE coupling for adherent biofilm to capture surface-specific dynamics.

6 MAP Parameter Estimates

Table 7: MAP θ for planktonic (28 params) and adherent (27 params).

Idx	Parameter	Name	Planktonic	Adherent
0	a_{11}	a(So-So)	+0.003	-0.791
1	a_{12}	a(So-An)	-1.179	-1.535
2	a_{22}	a(An-An)	+0.865	-2.828
3	a_{13}	a(So-Aa)	-0.888	+0.991
4	a_{23}	a(An-Aa)	+1.108	+2.474
5	a_{33}	a(Aa-Aa)	+2.400	+2.998
6	a_{14}	a(So-Vp)	-1.376	-0.410
7	a_{24}	a(An-Vp)	+0.382	+1.949
8	a_{34}	a(Aa-Vp)	+0.774	-0.823
9	a_{44}	a(Vp-Vp)	+1.359	+2.559
10	a_{15}	a(So-Fn)	+0.235	-2.409
11	a_{25}	a(An-Fn)	+0.053	+2.251
12	a_{35}	a(Aa-Fn)	+0.122	+2.848
13	a_{45}	a(Vp-Fn)	+0.067	-0.744
14	a_{55}	a(Fn-Fn)	+1.692	+2.213
15	a_{16}	a(So-Pg)	+1.053	-2.669
16	a_{26}	a(An-Pg)	-1.033	-0.609
17	a_{36}	a(Aa-Pg)	-0.869	+0.969
18	a_{46}	a(Vp-Pg)	-1.644	+2.089
19	a_{56}	a(Fn-Pg)	+5.468	+4.404
20	a_{66}	a(Pg-Pg)	+4.129	+3.874
21	μ_1	$\mu(\text{So})$	+3.683	+4.003
22	μ_2	$\mu(\text{An})$	+3.939	+2.523
23	μ_3	$\mu(\text{Aa})$	+2.993	+3.321
24	μ_4	$\mu(\text{Vp})$	+5.877	+2.222
25	μ_5	$\mu(\text{Fn})$	+0.000	+0.013
26	μ_6	$\mu(\text{Pg})$	+0.087	+0.037
27	K_{hill}	K_{hill}	+0.115	—

References

References

- [1] Siddiqui DA, Fidai AB, Natarajan SG, Rodrigues DC. Succession of oral bacterial colonizers on dental implant materials: an in vitro biofilm model. *Dental Materials* 38(2):384–396, 2022. DOI: 10.1016/j.dental.2021.12.021.

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